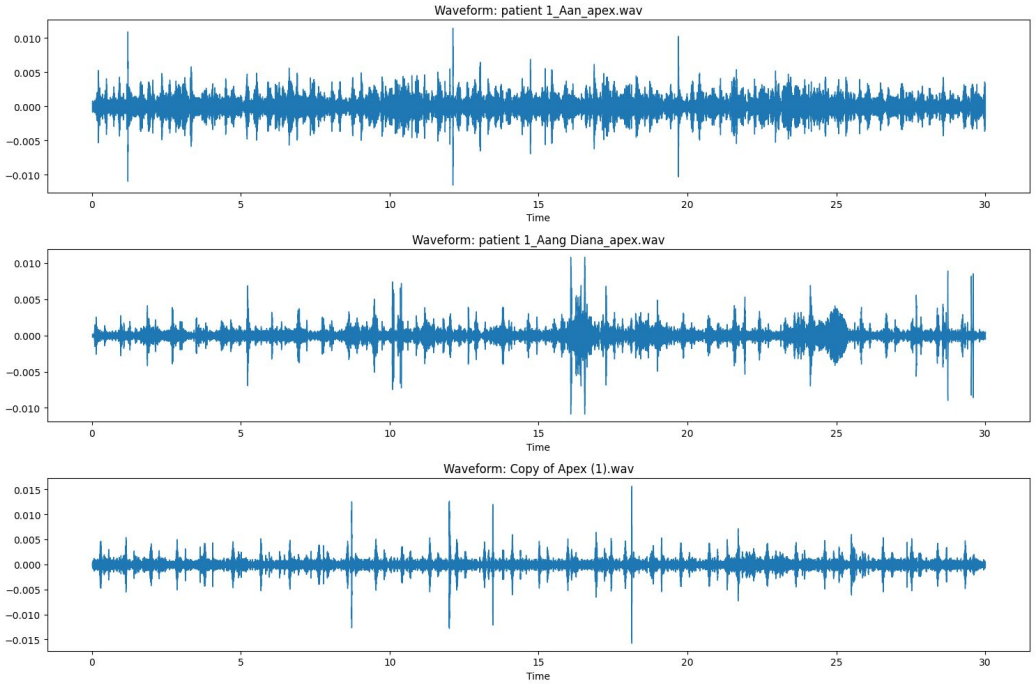
Phonocardiogram Visual Explanation

Randy Rizky Akram

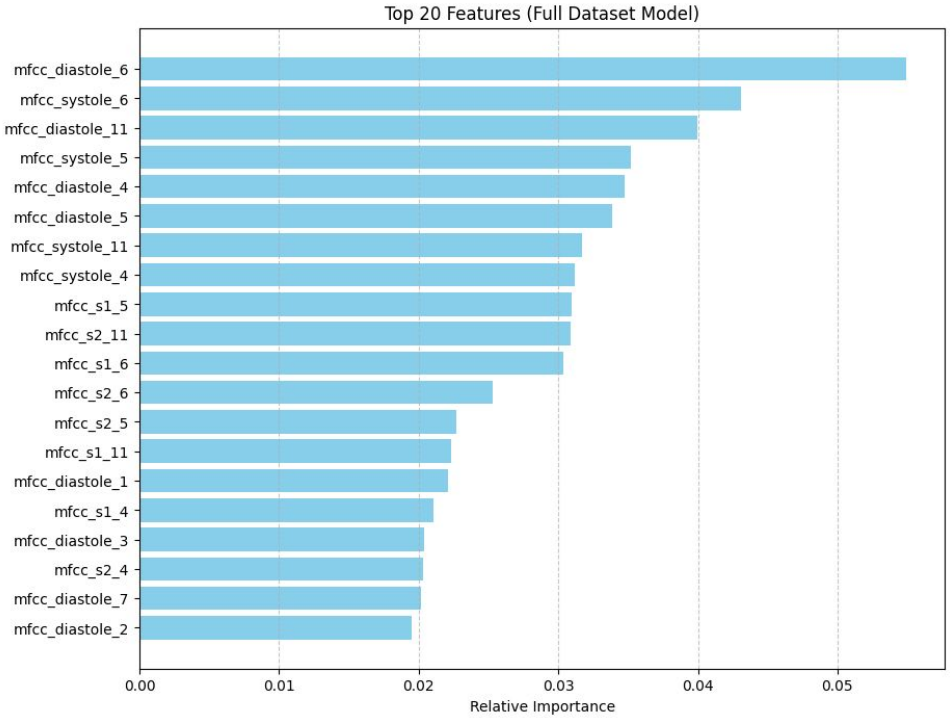
Data Input

- 4 Different recording location
  - Right Upper Sternal Border (RUSB) / Aortic
  - Left Upper Sternal Border (LUSB) / Pulmonic
  - Left Lower Sternal Border (LLSB) / Tricuspid
  - Apex / Mitral
- 30 Seconds per record
- 70 Patients for each Class
  - 4 Recording per Patients

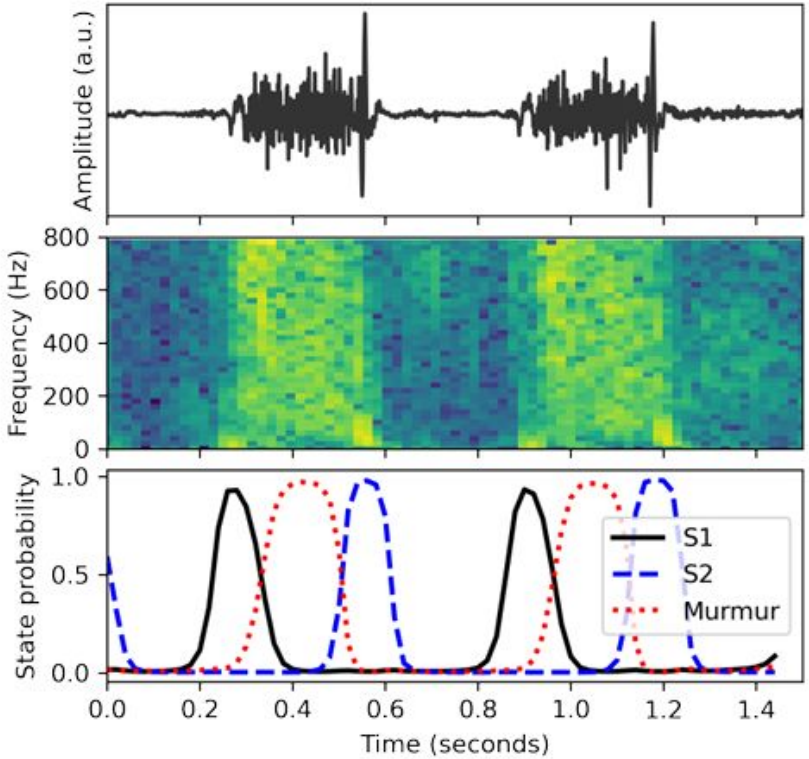


Machine Learning for Comparison

- Case Example: STEMI vs Normal
- Random Forest
- Accuracy 84%



Deep Learning Baseline

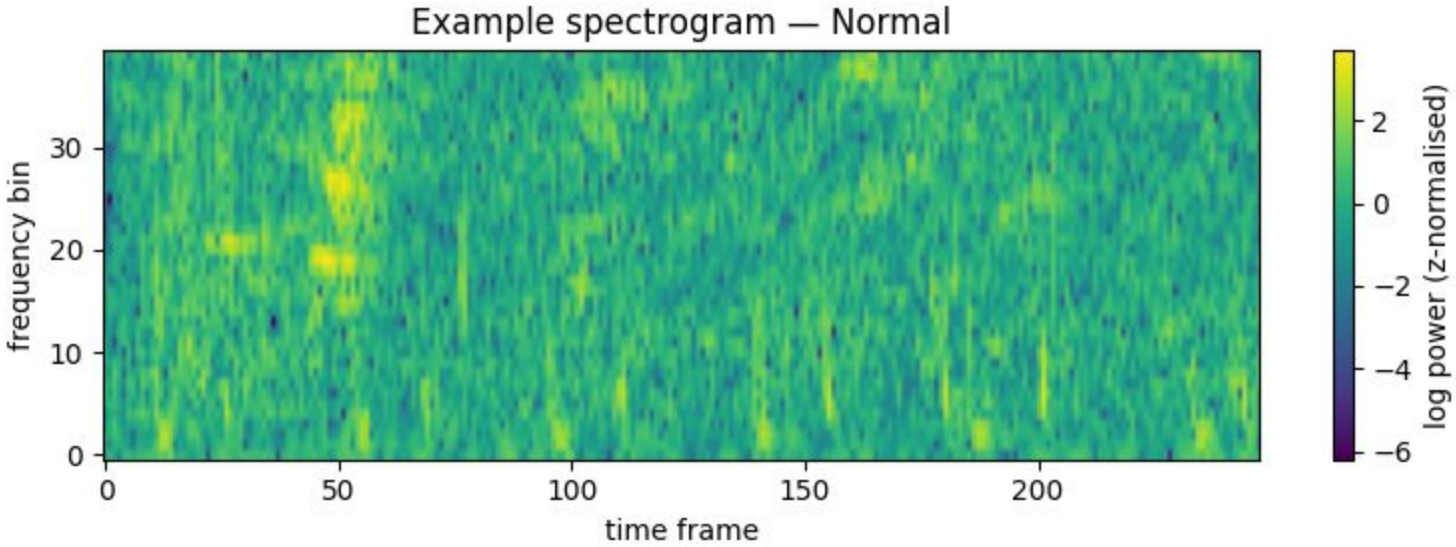


Parameter	Value
GRU hidden size	60
Number of GRU layers	3
GRU layer dropout	0.1
Fully-connected dropout	0.1
Fully-connected hidden sizes	[60, 40]

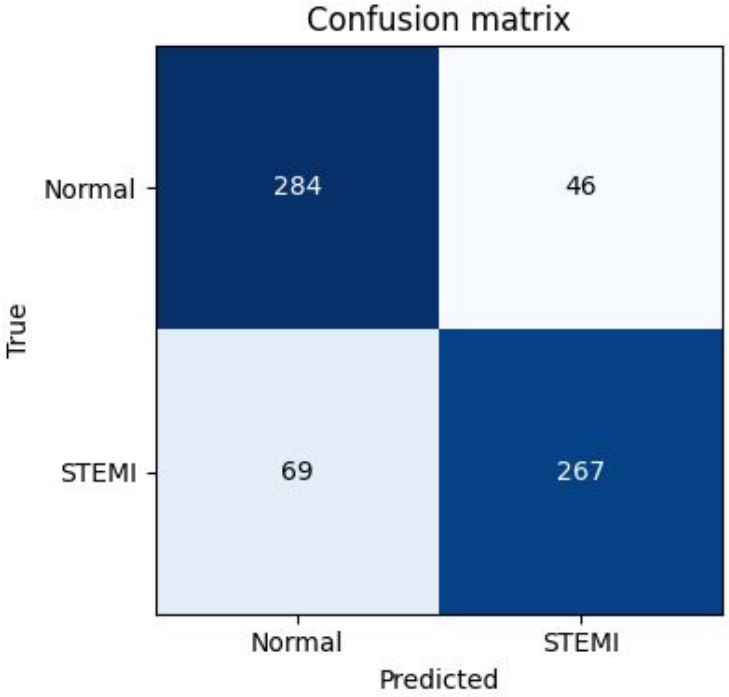
Proposed Deep Learning

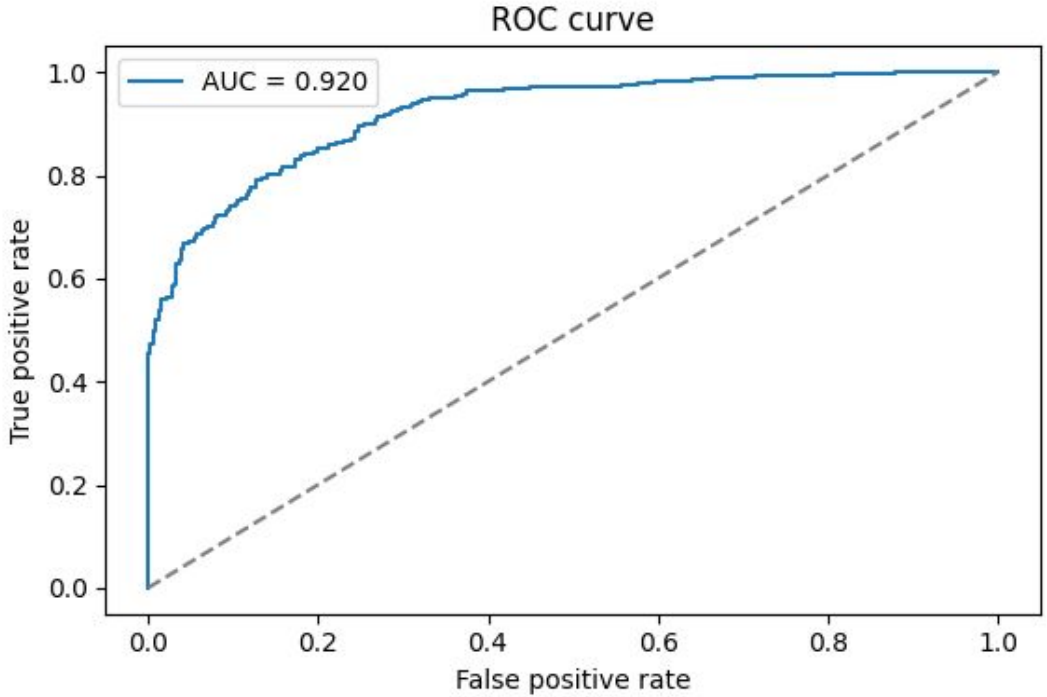
Step	Model	Details
0 (baseline)	RNN	GRU on raw spectogram
1	CRNN	+Conv2D front end for GradCAM
2	CRNN+Attention	+Learned attentioned pooling instead of mean-pooling

Preprocessing

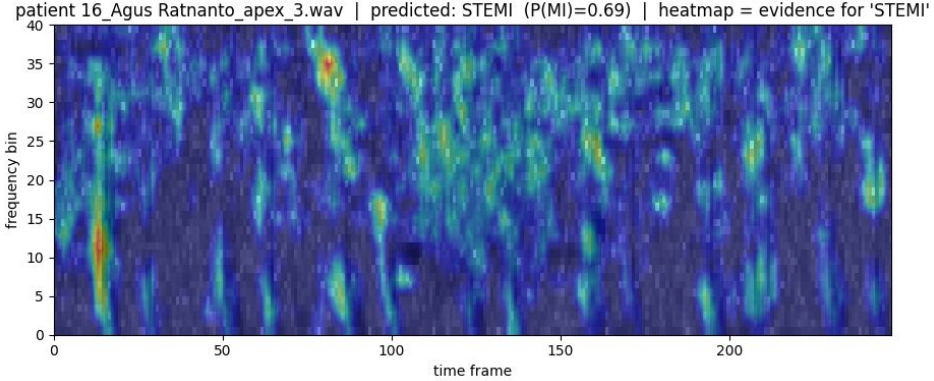


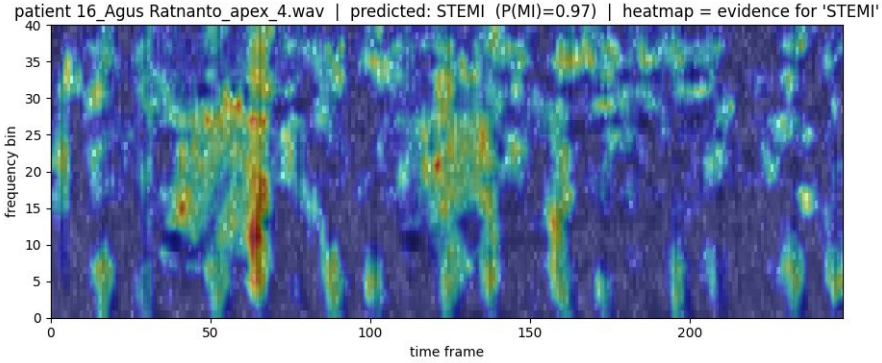
Result Example

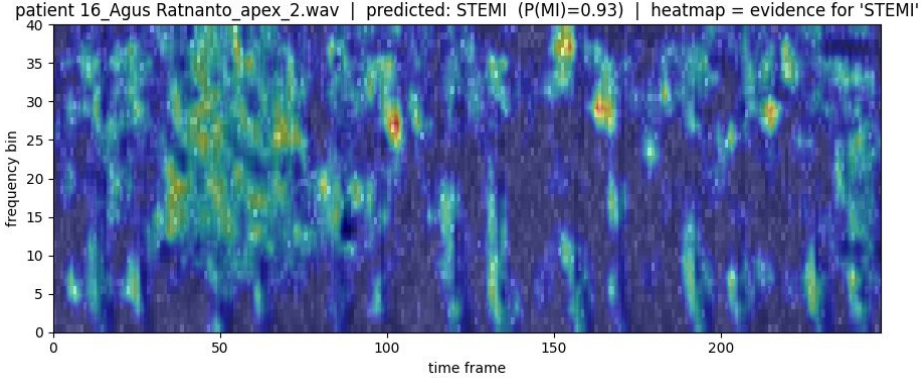


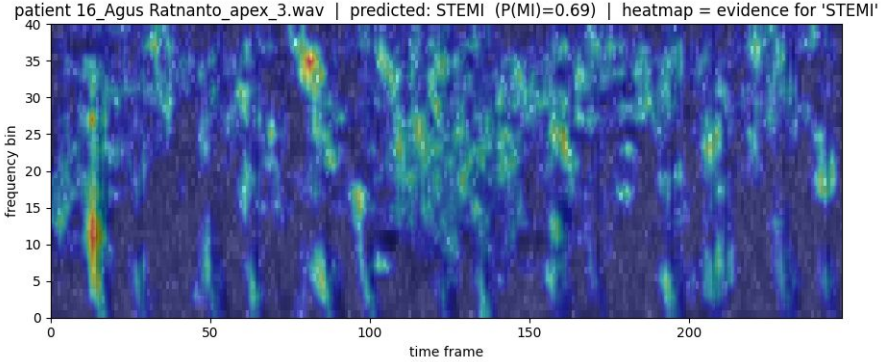


GradCAM STEMI

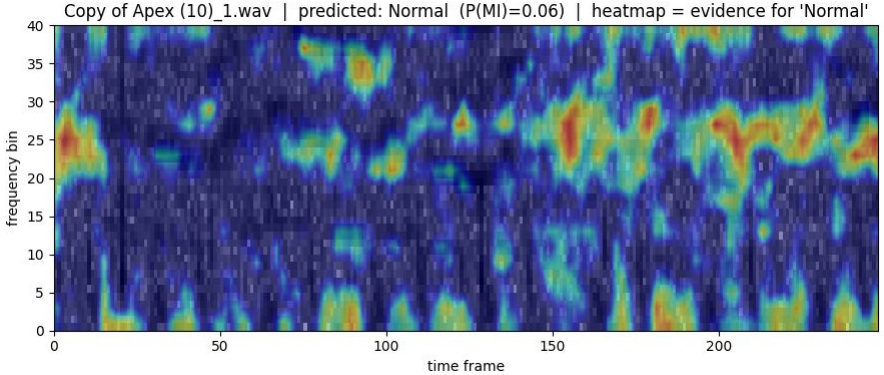


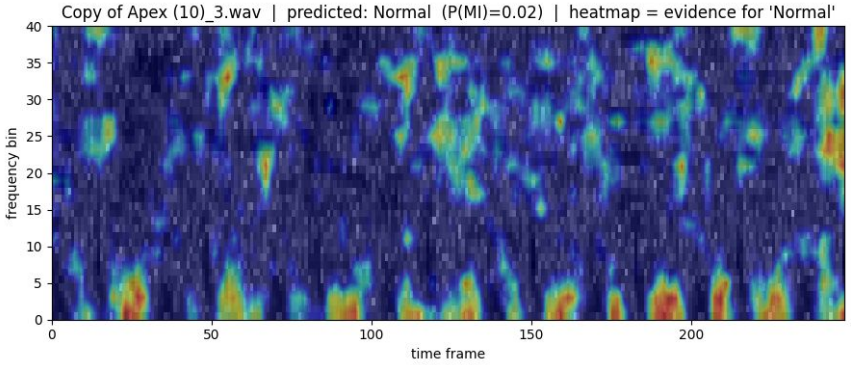


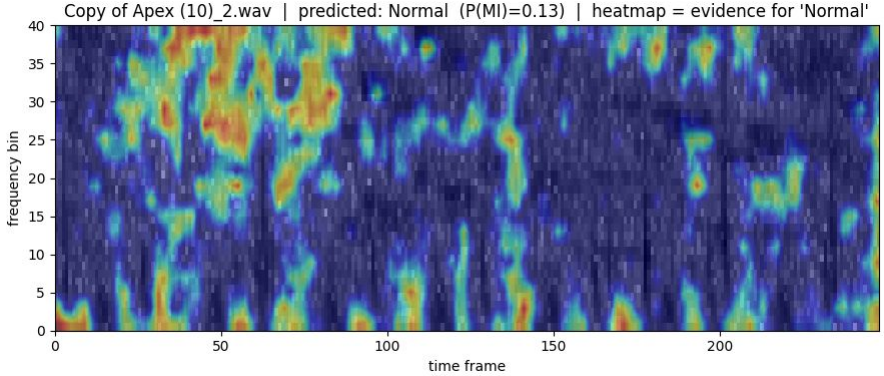


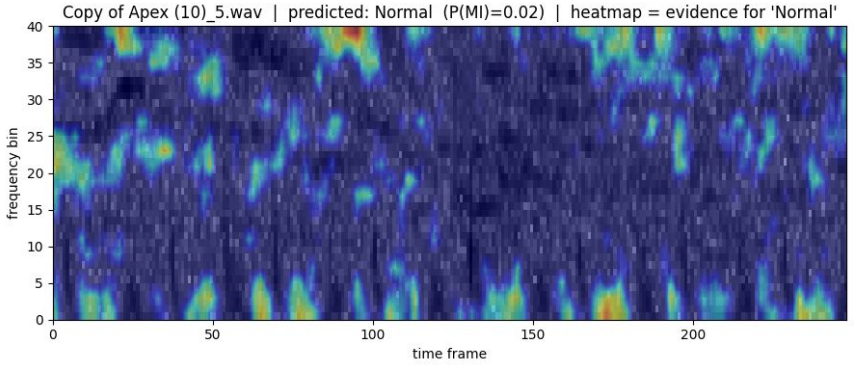


GradCAM Normal

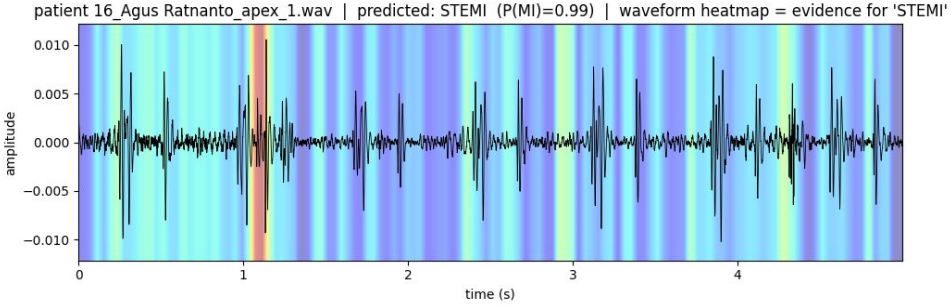


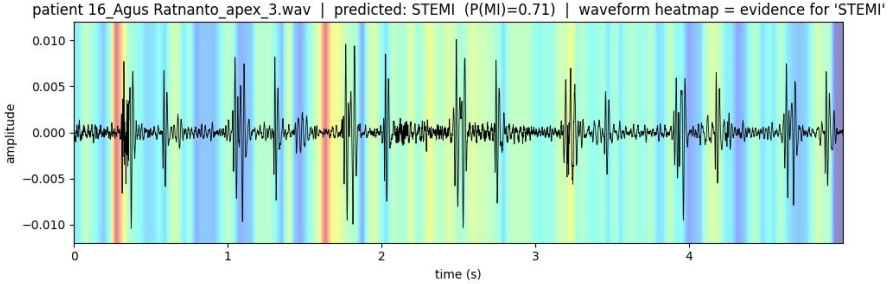


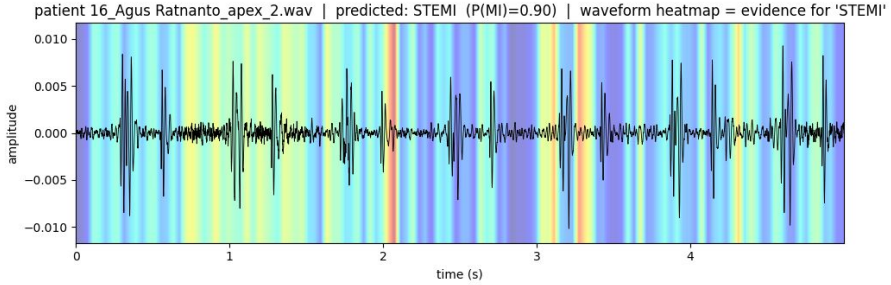


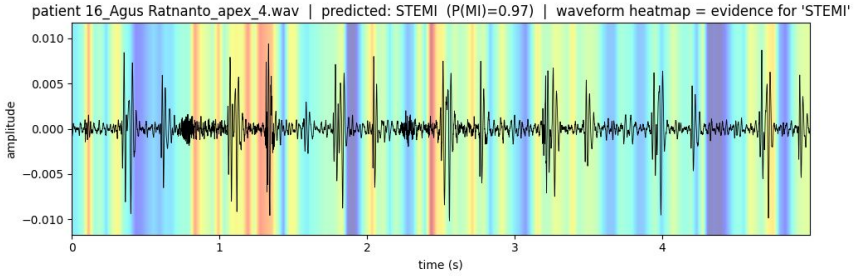


Projected on Waveform STEMI









Projected on Waveform Normal

